

Experiment 8

Implement the K-Nearest Neighbors (KNN) algorithm for image classification using the **Breast Cancer Dataset**. Experiment with different values of K and analyze their impact on model performance.

Tasks:

- Load and preprocess the dataset.
- Implement KNN for binary class classification.
- Experiment with different values of K and evaluate performance.
- Evaluate the model using Precision, recall, F1- Score, Accuracy.

Procedure

Step 1: Import Required Libraries

Step 2: Load the Breast Cancer Dataset using Scikit learn

Step 3: Separate Features and Target

- Input features $X \rightarrow 30$ numerical attributes.
- Target $Y \rightarrow$ binary class label

Step 4: Split the Dataset

Divide the dataset into training and testing datasets.

Step 5: Create the KNN Classifier

Create a KNN classifier using Scikit-learn. Initially select: $K=3$

Use Euclidean distance as the distance metric.

Step 6. Train KNN Model

Step 7. Predict Test Samples

Step 8: Calculate Evaluation metric Precision, Recall , Accuracy and confusion matrix

Step 9: Repeat the above procedure for different values of K, such as: $K=3,5,7,9$ and display comparison table

Step 10. Plot the Effect of K

Plot:

- K on the X-axis
- Accuracy on the Y-axis

Experiment 9

Implement a Decision Tree classifier using the ID3 algorithm to segment customers based on their purchasing behavior using the Online Retail dataset. Analyze the tree structure and discuss the feature importance.

Tasks:

- Load and preprocess the Online Retail dataset.
- Implement Decision Tree using the ID3 algorithm.
- Visualize the decision tree and analyze feature importance.
- Discuss how the tree structure helps in understanding customer behavior.

Aim

Customer Segmentation using Decision Tree (ID3) on Online Retail Dataset

Procedure

1. Load dataset
(<https://archive.ics.uci.edu/ml/datasets/Online+Retail>)
2. Clean data (Remove missing , negative and invalid values)
3. Create features
4. Define target
5. Train ID3 model
6. Visualize tree
7. Analyze feature importance